Shivang Ahlawat

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Education

Georgia Institute of Technology | Atlanta, GA Expected Graduation: *May 2028*
B.S. in Electrical Engineering, Minor in Mathematics *GPA: 3.83/4.00*

Experience

Georgia Institute of Technology Research *May 2026 - August 2026*
Undergraduate Researcher Metz, France

- Improved micro-LED cochlear implant thermal simulation in COMSOL Multiphysics by investigating 100-second time-dependent heat generation, tuned LED duty cycles, and power operation in biological tissue.
- Developed 1-2D finite-difference time-domain simulations to model transient heat propagation and in cochlear implants.
- Designed pulsed-heat source approach using duty-cycle fractions and pulse periods to model LED switching behavior.
- Evaluated simulation reliability through time-step comparisons examining constraints and the accuracy of thermal responses
- Reviewed existing biomedical engineering literature on cochlear implant thermal safety requirements, focusing on the 3 degree (Celsius) tissue-heating limit and temperature evaluation at local hotspots.

Institut Lafayette *May 2026 - August 2026*
Laboratory Assistant Metz, France

- Fabricated and characterized GaN-on-Si semiconductor prototypes using silicon wafers, photolithography, and Plasma-Enhanced Chemical Vapor Deposition (PECVD) SiNx passivation layers across a 2-25 micrometer range.
- Assisted with gas sensor microfabrication and semiconductor process evaluation, performing microscopic inspection and investigating potential causes of surface defects.
- Applied ellipsometry and profilometry to measure and analyze film thickness and surface uniformity across a 50–500 nm range.
- Conducted wafer-level defect and yield analysis using oblique dark-field inspection within a Class 100 cleanroom environment.

The Urban Indian *June 2024 – August 2025*
Logistics Manager Alpharetta, GA

- Managed daily fulfillment operations by coordinating the packing, quality checking, and delivery of up to 100+ orders.
- Developed an inventory management workflow that organized and tracked high-volume drops of over 500 individual garments.
- Coordinated with 3+ local delivery services, ensuring timely order pickup, tracking accuracy, and consistent transport reliability.

Projects

LED Ceiling Lights | Georgia Tech *August 2025 - March 2026*

- Designed and installed an ESP32-controlled lighting system driving 175+ feet of RGB LED strips across 8 lanes in a chapter room.
- Engineered a DC power distribution system using a fuse block and buck converters, stepping 24V supplies down to 5V to power the LED lanes and data hardware simultaneously
- Designed a data control circuit pairing the ESP32 (3.3V logic) with a logic level shifter to drive 5V LED data lines, with fuse protection between the central supply and each LED run
- Prototyped and validated the circuit with bench instrumentation (logic analyzer) to detect faults, resolved signal reflection issues through the addition of termination resistors and routing each lane through a dedicated buffered output on the level shifter.

Wearable ECG Monitor | Georgia Tech *August 2026 - Present*

- Working on an ECG circuit using an AD8232 analog front end and ESP32, sampling at 250 Hz via a timer interrupt.
- Implementing a C++ digital filter with a 0.5 Hz high-pass and 40 Hz low-pass to remove baseline drift, muscle noise, and 60-Hz power-line interference
- Eventually designing a wearable PCB prototype to move the breadboard circuit into a compact form that is wearable.
- Aiming to then fabricate and validate the board and compare its signal quality against the breadboard design.

Leadership

Mock Trial | Treasurer *August 2025 – Present*

- Manage funds and create budgets for up to $10,000-$15,000 throughout the year, closing attorney.
- Analyze affidavits and exhibits, create cross examinations and speeches, argue rules of evidence, regional and state qualifiers.

Skills

Programming/Software: Java, Python, C, C++, RISC-V Assembly, MATLAB, Linux, SQL, SolidWorks, LTspice, Visual Studio, Onshape, Microsoft Office, Copilot, Excel
Hardware & Embedded Systems: ESP32 microcontroller, Arduino Uno, Oscilloscopes, Logic Analyzers, NI myDaq
Circuit & PCB Design: KiCad, Schematic Capture, PCB Layout, Power Regulation, Logic Level Shifting
Digital Signal Processing: Digital Sampling, Filter Design, Spectral Analysis, Discrete Fourier Transform, Z-Transform
Circuit Analysis: Thevenin-Norton Equivalent circuits, DC and AC Steady-State analysis, Transformers, Filters
Digital System Design: Boolean logic, Combinational and Sequential Logic, Mealy and Moore State Machines, Data Paths
Mathematics: Differential Equations, Linear Algebra, Quantum Computation, Vector Calculus, Proofs, Probability & Statistics